

PEDRO LOURES · UNITY DEVELOPER · TORONTO

# Pedro Loures

I turn game designs into systems that ship.

Unity developer since 2012: gameplay systems, VR interaction, editor and Blender tools, networking and data pipelines, for studios, schools and clients like Fiat and Vale. A Vale presentation system and a Blender animation add-on I built are still in use. Most recently I was Unity lead on Anchored Memories, a full-body haptic VR experience at UT Austin.

Unity · C#

Gameplay systems

Tools · Python

Networking & data

VR / Haptics

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## SELECTED PROJECTS



HAPTIC VR · UNITY · 2026

### Anchored Memories

Five-act haptic VR story on climate migration · Unity lead and project engineer · UT Austin, 2026

**Role** Unity Lead · Project Engineer (D2R Studios, volunteer) · **Year** 2026 · **Engine** Unity · **Team** 12+ (TXI, D2R Studios, UT Austin advisors) · **For** Texas Immersive Institute, UT Austin

- Engineered the project's foundation: the barebones architecture, VR base interactions and audio base code the rest of the team builds on
- Coded the bridge that brings the haptics designer's patterns into Unity for the bHaptics TactSuit X40
- Built all five acts as separate, ready-to-work spaces, each with its own environment prepared for the ocean changes
- Shaders, texturing, rain particle effects with optimization tests, and importing and configuring assets, including the ocean, lent from my own Asset Store library where licenses allowed
- Now tuning the release build for the best settings without direct access to the target hardware, testing through rounds of builds with the team

#### LEADERSHIP

- Moved the project from Unreal to Unity when I joined, so a small, part-time team could reach a working build for the July 2026 showcase
- Main developer on the levels for more than half of the project, and on call as backup for the whole year
- Set up and maintained the GitHub project board, supported the Scrum Master with planning and task breakdown, and onboarded the students into the project



MANAGEMENT SIM · UNITY · 2021

## Dream Arcade Simulator

Arcade management sim · Centennial College capstone, team of 5

**Role** Gameplay Programmer · **Year** 2021 · **Engine** Unity · **Team** 5 · **For** Centennial College capstone

- Day/night cycle with skybox, exposure and light blending
- The whole night phase: shop with tabs, grid placement, move, rotate and sell
- Audio manager with fade-in/out jobs, and a data-driven cheat code system using UnityEvents
- Fixed AI code that overloaded the CPU (stutter, FPS drops and overheating) because objects weren't being disposed



VR · 4 MINIGAMES · SOLE PROGRAMMER

## VR Carnival

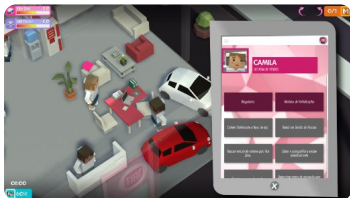
Four seated VR minigames in under a month · sole programmer

**Role** Sole Programmer, Game Design · **Engine** Unity · SteamVR · **Team** 3 · **For** Centennial College VR course

- MiniGameManager singleton base class with overridable lifecycle hooks, so only one minigame runs at a time
- Dissolve shader effect for materializing the bow and hammer in the player's hand
- Extended SteamVR's interaction samples (grabbing, bow, buttons) into four distinct games

### LEADERSHIP

- On a short-handed team, took on all the programming and scoped the project to four core games in under a month, polishing only after all four were built



TRAINING SIM · CLIENT: FIAT

## Fiat Quality Manager Training

Data-driven training simulation for Fiat dealership Quality Managers

**Role** Dev Lead · Programmer · Game Design team · **Engine** Unity · **For** Fiat · D2R Studios

- Content pipeline: Fiat's Excel scenario sheets → JSON → NoSQL database, downloaded by Unity at runtime
- 76 real-world scenarios at launch, with a web admin so the client can add more without a new build

### LEADERSHIP

- Dev lead and one of the main programmers, while also designing the game as part of a team where everyone helped shape it



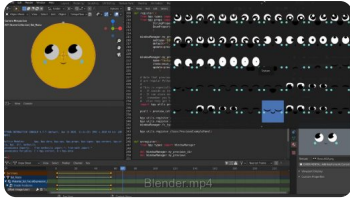
MULTI-SCREEN APP · CLIENT: VALE · STILL IN USE

## Vale · Porto de Tubarão

Three-screen presentation system controlled from a tablet · 2012 → 2019

**Role** Desktop developer (2012) · Rebuild developer (2019) · **Year** 2012 / 2015 / 2019 · **Engine** Unity · **For** Vale · Zor Design · D2R Studios

- Custom sprite system written before Unity had native 2D
- Content pipeline through the Resources folder: new images only need the right file names, with no code or level changes
- In one month for the 2019 version: rebuilt the tablet app from scratch, moved the PC app off OnGUI and added real networking with UNet
- Still in use today, more than a decade after the first version



ANIMATION TOOL · PYTHON · STILL IN USE

## Blender Expressions Add-on

Python add-on that speeds up facial-expression keyframing · built for D2R Studios' animator, 2019

**Role** Tool designer · Programmer · **Year** 2019 · **Engine** Blender · Python · **For** D2R Studios

- Lists every facial texture for a character's face parts, read from a folder naming convention, so animators pick expressions from one panel
- Saves around 3 clicks per keyframe when swapping faces, plus a one-click keyframe button
- Built as a side project alongside my regular work, fitted into the studio's existing animation pipeline; the animator it was made for still uses it



HORROR · GAME JAM · TEAM LEAD

## Coma · College Game Jam

Horror game jam on the theme "Another World" · led a team of newer developers

**Role** Team Lead · Programmer · **Year** 2021 · **Engine** Unity · **For** College midterm game jam

- Procedural maze generator (adapted from Kodeco's maze tutorial) whose output was saved and then decorated by hand
- Reusable trigger system built on UnityEvents, with cooldowns and one-shot options

### LEADERSHIP

- Led a team of newer students (placed there by the coordinator to lead) and set up a prefab workflow so the artists could do the level design themselves



EDUCATIONAL · LUA · UNITY

## Mitologia da Programação

Educational game where players solve Greek-myth challenges by writing Lua

**Role** Game Designer · Level Designer · Programmer · **Engine** Unity · embedded Lua · **Team** 2

- Embedded a Lua interpreter in Unity, so players' code compiles and runs inside the game
- Checks each solution against the expected output, with custom hint messages or the compiler's own errors



PUZZLE PLATFORMER · SBGAMES AWARD · 2013

## Loneliness

Shadow-puzzle platformer · SBGames 2013: 3rd in Popular Vote + Best Short Paper

**Role** Game Designer · Level Designer · Programmer · **Year** 2013 · **For** PUC capstone

- Shadow replay system: moved from frame-data recording to input recording to fix desyncs on frame drops (the postmortem covers what that late rewrite cost)

## MORE WORK

Animal Defense, Fiat Personalização, Custom Defense, Procedural Dungeon Generator, Chaveirinho, Digital Wonderland, Recifuturo. Details, videos and code at [pedroloures.com/projects](https://pedroloures.com/projects).

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